

## Lab 1 — Requirements Table

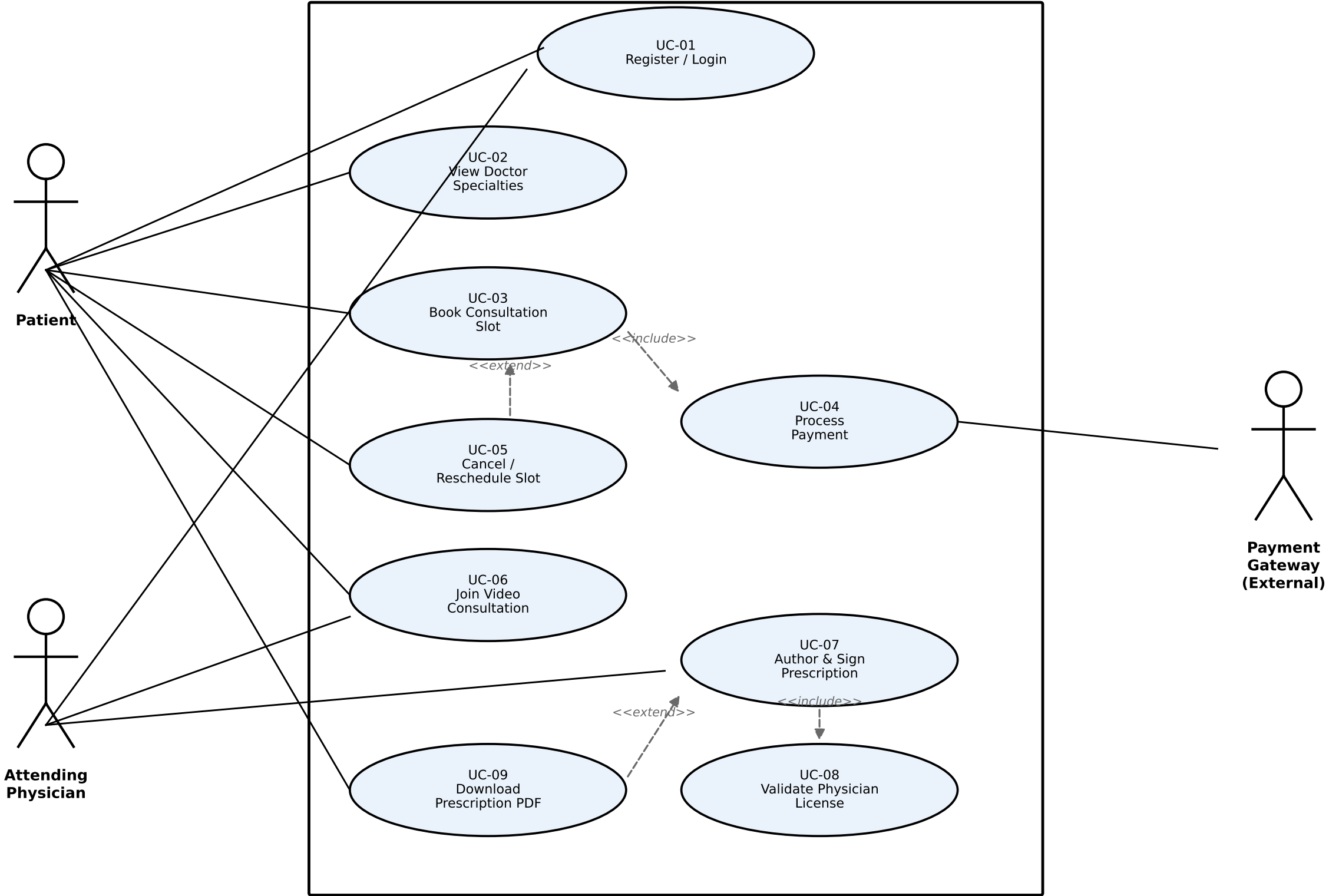
### Problem Statement #11: Telemedicine Slot Booking & Prescription Portal

Actors: Patient, Attending Physician, Payment Gateway (external system)

| Req ID         | Type          | Description                                                                                                                                                                                    | Priority | Acceptance Criteria                                                                                                                                                                                                   | Rationale                                                                                                    |
|----------------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>FR-001</b>  | Functional    | The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.                                        | High     | Pass: Valid cryptographic signature attached to prescription PDF. Fail: Prescription generated without doctor license validation.                                                                                     | Ensures prescriptions are authentic, legally traceable, and tamper-evident.                                  |
| <b>FR-002</b>  | Functional    | The system shall allow a patient to view a list of doctor specialties along with each doctor's available consultation slots for the next 7 days.                                               | High     | Pass: Selecting a specialty displays only doctors of that specialty with real-time slot availability. Fail: Booked/expired slots are shown as available.                                                              | Patients need accurate, up-to-date information to choose a suitable doctor and time.                         |
| <b>FR-003</b>  | Functional    | The system shall allow a patient to book a video consultation slot only after successful payment confirmation, and shall prevent the same slot from being booked by more than one patient.     | High     | Pass: Concurrent booking attempts on the same slot result in exactly one confirmed booking and one 'slot unavailable' message. Fail: Two patients are confirmed for the same slot.                                    | Prevents double-booking and revenue loss, and guarantees a 1:1 doctor-patient slot mapping.                  |
| <b>FR-004</b>  | Functional    | The system shall automatically generate and send an encrypted, single-use tele-consultation room link to both patient and physician 10 minutes before the scheduled slot time.                 | Medium   | Pass: Link is delivered to both parties within $10 \pm 1$ minutes of slot start and expires after the session ends. Fail: Link is reusable or delivered late.                                                         | Protects consultation privacy and ensures both parties can join on time.                                     |
| <b>FR-005</b>  | Functional    | The system shall allow a patient to view and download the digitally signed prescription PDF from their consultation history after the consultation is marked complete.                         | Medium   | Pass: Download is available only after consultation status = 'Completed' and prescription status = 'Signed'. Fail: Unsigned or draft prescriptions can be downloaded.                                                 | Gives patients timely, verifiable access to their medical records while preventing premature disclosure.     |
| <b>NFR-001</b> | Nonfunctional | All video session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest with AES-256.                                                                     | High     | Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.                                                                                                                     | Protects sensitive patient health data (PHI) in compliance with data-protection regulations.                 |
| <b>NFR-002</b> | Nonfunctional | The portal shall maintain 99.5% uptime measured monthly (excluding scheduled maintenance windows) and support at least 500 concurrent video consultations without degradation in call quality. | High     | Pass: Monthly uptime logs show $\geq 99.5\%$ availability and load tests at 500 concurrent sessions show no drop in audio/video quality ( $< 150$ ms jitter). Fail: Uptime $< 99.5\%$ or quality degrades under load. | Healthcare consultations are time-critical; unavailability or poor call quality directly harms patient care. |

Shaded rows (FR-001, NFR-001) were provided in the lab handout; all other rows were authored for this submission.

Telemedicine Slot Booking & Prescription Portal



— Association  
- - - <<include>> / <<extend>> (dependency)

# Use-Case Flow Specification

*Problem Statement #11: Telemedicine Slot Booking & Prescription Portal*

## Use Case: UC-03 — Book Video Consultation Slot

### Actors

Primary: Patient | Secondary: Payment Gateway (external system)

### Preconditions

- Patient is registered and logged into the portal.
- Patient has already viewed doctor specialties and selected a physician (UC-02).
- The selected physician has at least one open (unbooked) slot in the next 7 days.

### Postconditions

- Success: The slot is marked 'Booked' against the patient, payment is confirmed, and a booking confirmation with an encrypted consultation link is scheduled for delivery.
- Failure: The slot remains 'Available' and no payment is captured.

### Main Success Scenario

1. Patient selects a physician and views that physician's open slots for the next 7 days.
2. Patient selects an available slot and clicks “Book Slot.”
3. System locks the slot temporarily and displays a booking summary (physician, date/time, consultation fee).
4. Patient confirms the booking and selects a payment method (card / UPI / wallet).
5. System invokes UC-04 (Process Payment) via an <<include>> relationship, forwarding the amount and patient details to the Payment Gateway.
6. Payment Gateway authorizes the transaction and returns a success response.
7. System permanently marks the slot as “Booked,” releases the temporary lock, and records the transaction ID.
8. System schedules generation of the encrypted tele-consultation room link (UC-04a) for 10 minutes before the slot time.
9. System displays a booking confirmation screen and emails/SMS a receipt to the patient.
10. Use case ends successfully.

### Alternate Flow — 8a. Payment Declined

- 8a1. Payment Gateway returns a “declined” response to the system.
- 8a2. System releases the temporary slot lock, making the slot available to other patients again.
- 8a3. System displays a “Payment Failed” message and prompts the patient to retry with the same or a different payment method.
- 8a4. If payment fails again after 2 attempts, the booking attempt is cancelled and the patient is notified to select a slot again; the use case ends in failure.